

Deliverable 4 – Data Quality Assessment Report

Healthcare Patient Experience Analytics using Power BI

Introduction

Data quality is a critical component of any Business Intelligence solution. Poor-quality data can lead to inaccurate reports, misleading insights, and poor business decisions. Before building the Power BI dashboards, the healthcare datasets were assessed and cleaned using **Power Query Editor** to ensure consistency, accuracy, and reliability.

This report summarizes the major data quality issues identified during data preparation, their potential business impact, and the corrective actions taken to improve the overall quality of the dataset. The assessment follows the data quality requirements outlined in the project brief.

Data Quality Assessment

Data Quality Issue	Business Impact	Resolution
Missing Values	Missing records can result in inaccurate KPI calculations and incomplete analysis.	Missing values were identified using Power Query and handled by replacing, filtering, or retaining them based on business requirements.
Duplicate Records	Duplicate records can inflate patient satisfaction percentages and distort performance analysis.	Duplicate rows were identified and removed using the Remove Duplicates feature in Power Query.
Invalid State Codes	Incorrect state codes can lead to inaccurate mapping and regional analysis.	State codes were validated against the States table and corrected or

Data Quality Issue	Business Impact	Resolution
Missing Survey Responses	Missing survey responses reduce the reliability of patient satisfaction analysis.	removed where necessary. Blank survey response values were identified and treated appropriately during data preparation.
Inconsistent Reporting Periods	Inconsistent reporting periods prevent accurate trend and year-over-year analysis.	Reporting period values were standardized to ensure consistency across all datasets.
Invalid Percentage Values	Percentage values below 0% or above 100% produce incorrect KPI calculations.	Percentage columns were validated and incorrect values were corrected or excluded from analysis.
Data Type Mismatches	Incorrect data types may cause calculation errors and relationship issues.	Data types were converted to appropriate formats such as Text, Whole Number, Decimal Number, or Date using Power Query.

Data Cleaning Process

The following data preparation activities were performed in **Power Query Editor** before loading the data into the Power BI data model:

- Reviewed all datasets for completeness and consistency.
- Identified missing and null values.
- Removed duplicate records.
- Verified state names and state codes.
- Standardized reporting period values.
- Validated percentage values.
- Corrected inconsistent data types.
- Renamed columns for better readability.
- Ensured consistency across all related tables.
- Loaded the cleaned datasets into the Power BI data model.

Power Query Transformations

The following Power Query transformations were used during data preparation:

Transformation	Purpose
Remove Duplicates	Eliminate duplicate records
Replace Values	Correct inconsistent values
Change Data Type	Assign appropriate data types
Rename Columns	Improve readability
Trim & Clean Text	Remove unnecessary spaces and hidden characters
Filter Rows	Remove invalid or unnecessary records
Remove Blank Rows	Improve data quality
Validate Percentage Values	Ensure values remain within acceptable ranges
Standardize Reporting Periods	Maintain consistency across datasets

Data Quality Summary

The data quality assessment ensured that:

- The datasets were complete and suitable for analysis.
 - Duplicate records were eliminated.
 - Invalid state information was corrected.
 - Percentage values were validated.
 - Reporting periods were standardized.
 - Data types were consistent across all datasets.
 - Relationships between tables could be created without errors.
 - The cleaned data supported accurate KPI calculations and reliable dashboard reporting.
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Business Benefits

Performing data quality assessment provides several business benefits:

- Improves the accuracy of dashboard visualizations.

- Ensures reliable KPI calculations.
 - Reduces reporting errors.
 - Supports accurate benchmarking and trend analysis.
 - Enhances confidence in business decisions.
 - Improves overall dashboard performance.
 - Provides consistent and trustworthy healthcare insights.
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Conclusion

The data quality assessment process ensured that all healthcare datasets were clean, consistent, and ready for analytical reporting. By identifying and resolving issues such as missing values, duplicate records, invalid state codes, inconsistent reporting periods, invalid percentage values, and data type mismatches, the project established a reliable foundation for data modeling, KPI development, and dashboard creation. The cleaned dataset supports accurate business insights and enables healthcare stakeholders to make informed, data-driven decisions.

